Assignment 3

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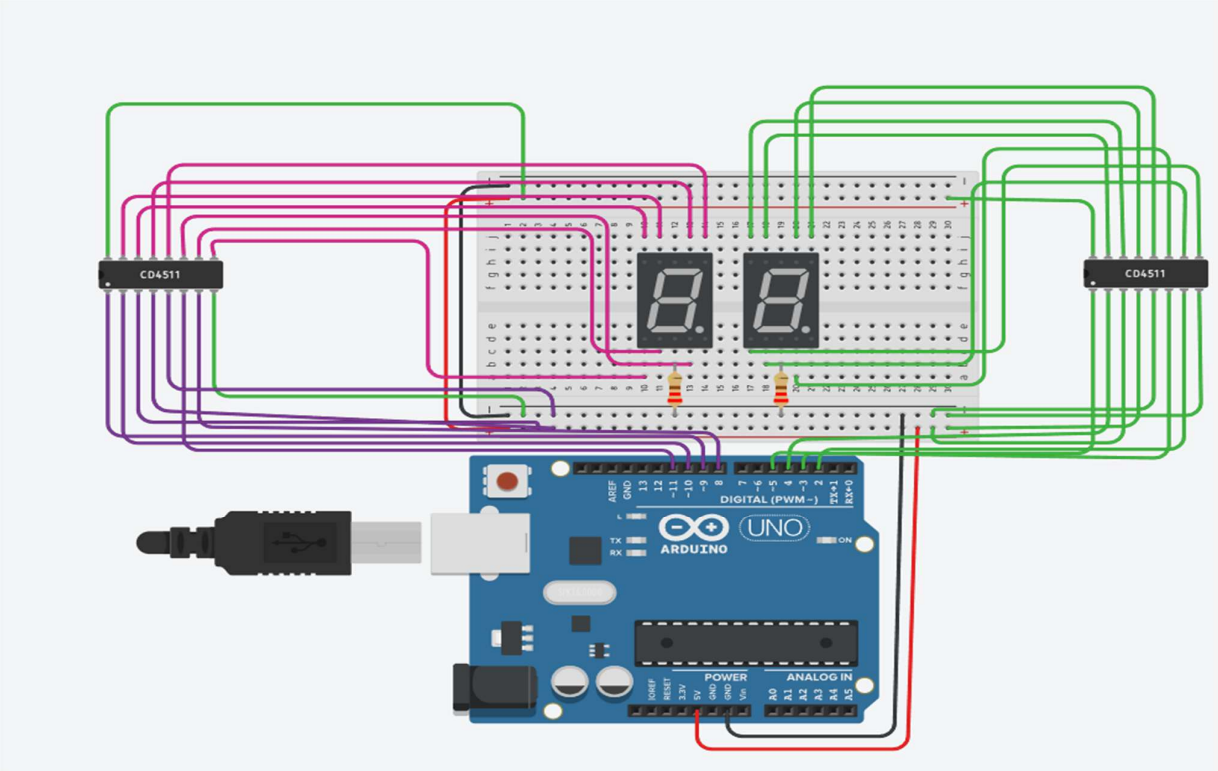
Design a digital counter that displays numbers from 0 to 30 using two 7 segment displays

Components List:

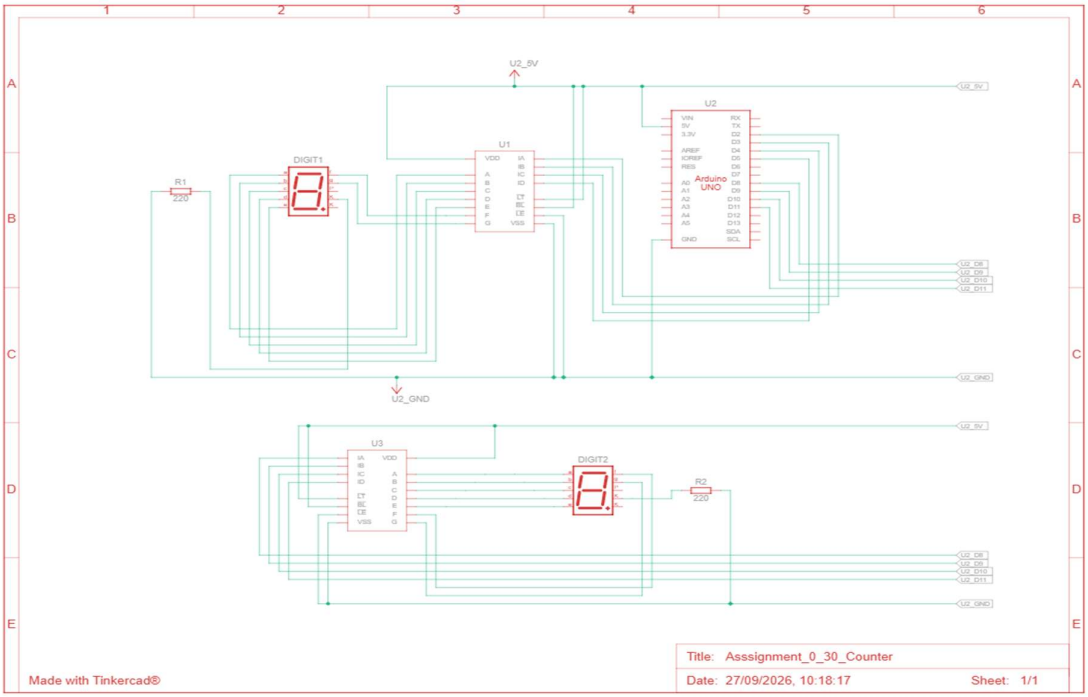
Component List

Name	Quantity	Component
U1 U3	2	7-Segment Decoder
1	1	Breadboard Small
U2	1	Arduino Uno R3
Digit1 Digit2	2	Cathode 7 Segment Display
R1 R2	2	220 Ω Resistor

Circuit Diagram:



Schematic:



CODE:

```
int ic0_pins[] = {2, 3, 4, 5};
int ic1_pins[] = {8, 9, 10, 11};
void setup(){
 for (int i = 0; i < 4; i++){
  pinMode(ic0_pins[i], OUTPUT);
  pinMode(ic1_pins[i], OUTPUT);
 }
}
void loop(){
 for (int num1 = 0; num1 <= 3; num1++){
  for (int j = 0; j < 4; j++){
   digitalWrite(ic1_pins[j], (num1 >> j) & 1);
  }
   for (int num0 = 0; num0 <= 9; num0++){
    for(int i = 0; i < 4; i++){
     digitalWrite(ic0_pins[i], (num0 >> i) & 1);
    }
    delay(1000);
    if ((num1 ==3) && (num0 == 0)){
     while(1);
    }}}}
```